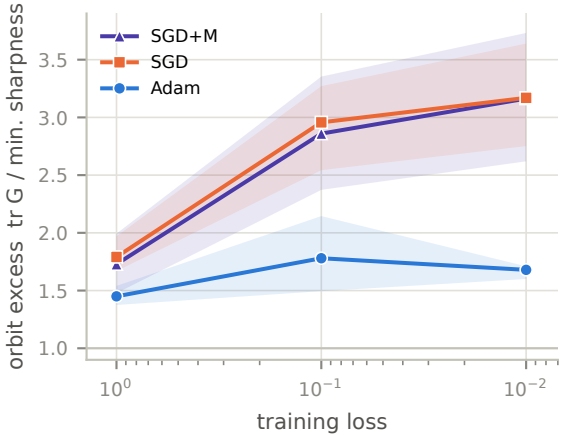
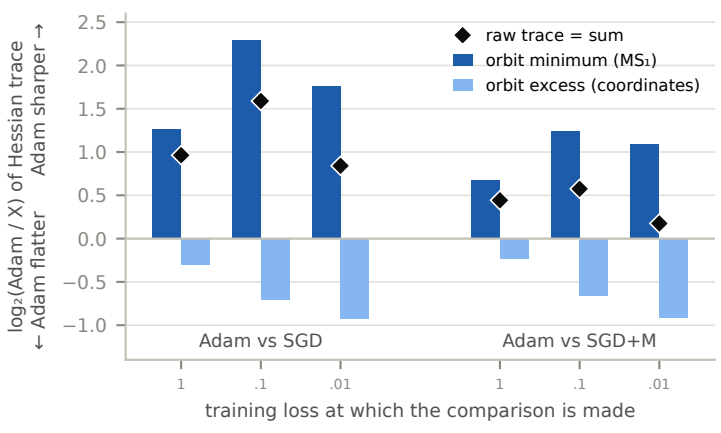

(a) Orbit excess (median, min–max)
SGD+M
SGD
Adam
orbit excess tr G / min. sharpness
3.5
3.0
2.5
2.0
1.5
1.0
10^{0}
10^{-1}
10^{-2}
training loss


(b) Raw trace = orbit minimum × orbit excess
raw trace = sum
orbit minimum (MS_1)
orbit excess (coordinates)
log₂(Adam / X) of Hessian trace
← Adam flatter Adam sharper →
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
Adam vs SGD
Adam vs SGD+M
1
.1
.01
1
.1
.01
training loss at which the comparison is made